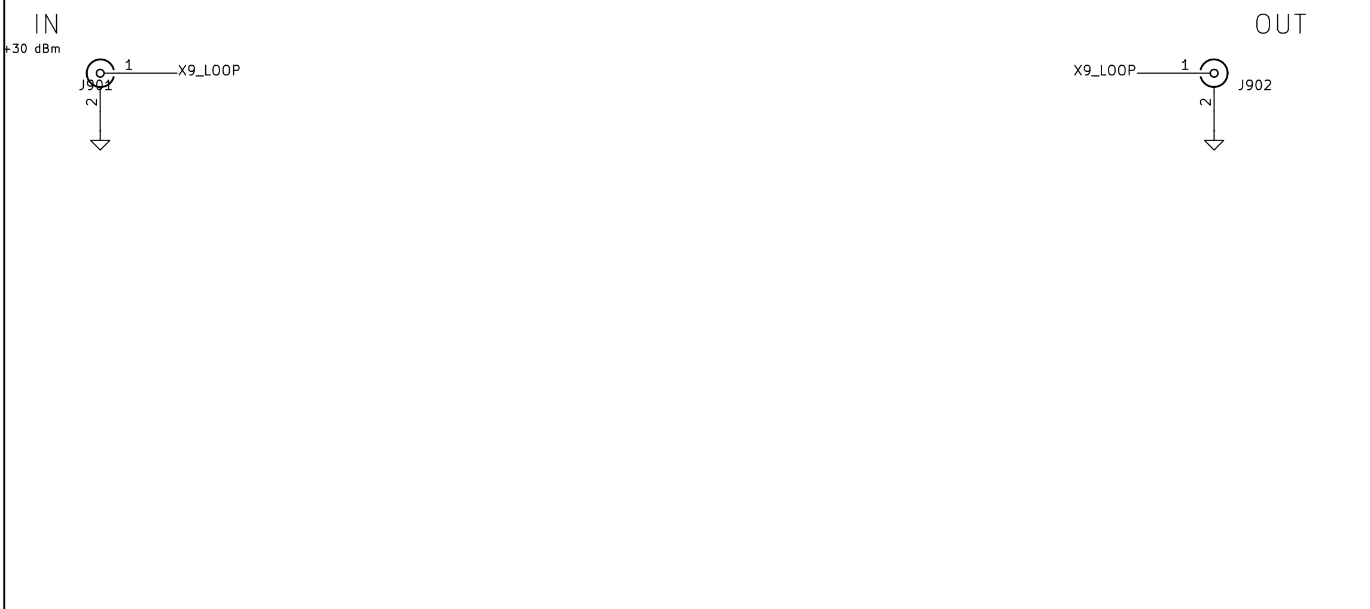
X9 ODMR loop

cell (0,0)

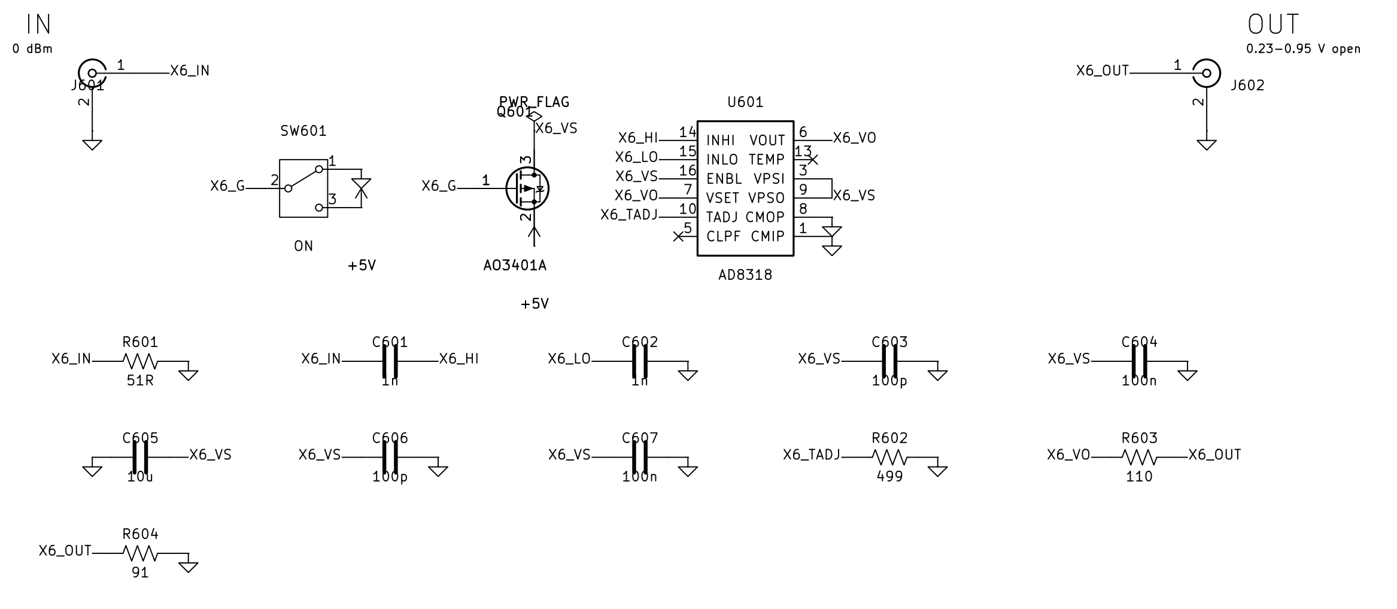


EXPECT: a 50 ohm through line (the loop adds about 1.5 nH); at its centre about 50 uT (0.5 G) of microwave field for +20 dBm through it into 50 ohm, a Rabi frequency near 1 MHz for NV centres; twice that with OUT shorted

A one–turn loop, 1.6 mm across (0.3 mm track), in series with a 50 ohm line between IN and OUT. B at its centre is mu0 I / 2a: 63 mA peak (+20 dBm into 50 ohm) gives 50 uT. No resonance, so no tuning: sweep 2.6–3.2 GHz for ODMR in any bias field. Copper is cleared on every layer within 2 mm of the loop. Lay the diamond on the loop, shine green on it from above and read the red with the ECP5 board's photodiode.

X6 log detector, AD8318

cell (1,0)

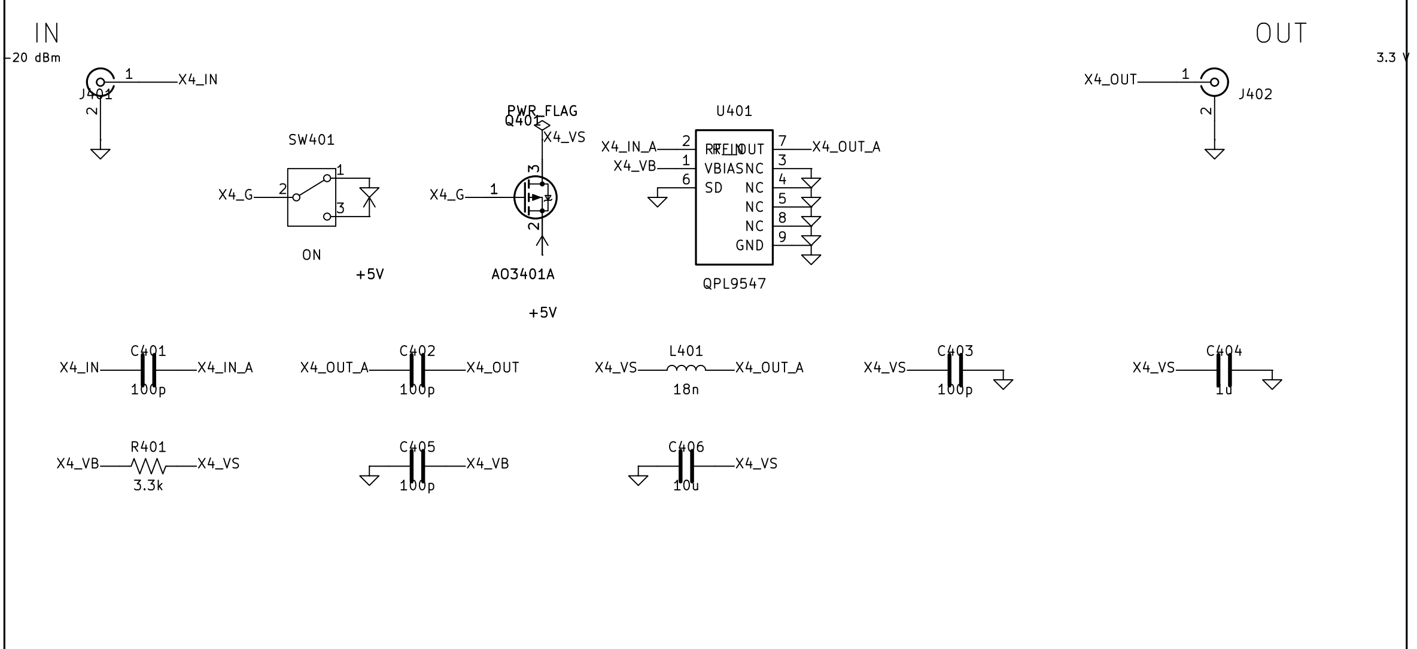


EXPECT: VOUT falls 25 mV per dB of input from –60 to 0 dBm, 1 MHz–8 GHz (2.1 V at –60 dBm, 0.6 V at 0 dBm at the chip); at the jack 0.45 x that from 50 ohm: 0.27–0.95 V into the ADC's high impedance, half into 50 ohm; 10 ns response

ADI AD8318: a nine–stage log amplifier with detectors, 60 dB of range to 8 GHz, temperature–compensated (499 ohm on TADJ suits 0.9–2.2 GHz). VSET = VOUT: –25 mV/dB with an intercept near +20 dBm. CLPF is open, so it follows a pulse in 10 ns: a radar's or a pulsed experiment's envelope at the ADC's 65 MS/s. The 110 / 91 ohm divider keeps the 0.5–2.1 V inside the ADC's +–1 V and makes the output a 50 ohm source. A power meter, a scalar network analyzer with X3, antenna patterns.

X4 LNA, QPL9547

cell (2,0)

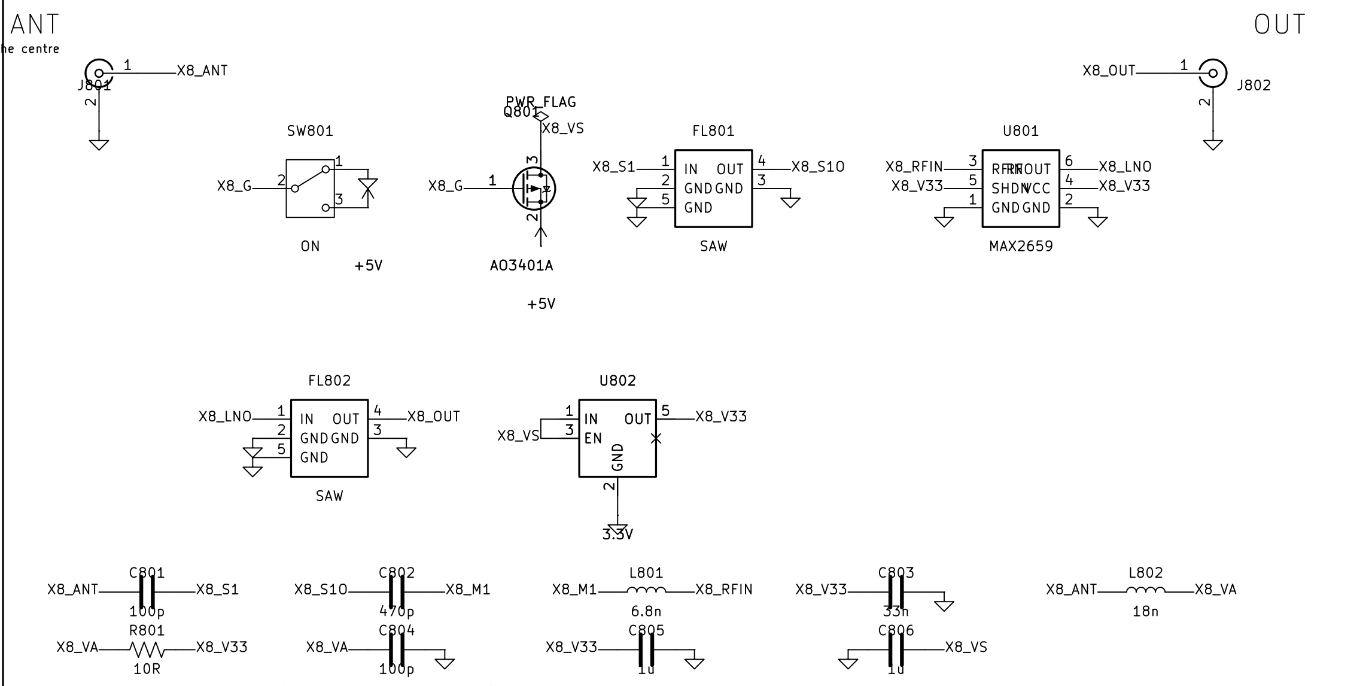


EXPECT: 19.5 dB of gain and a 0.3–0.5 dB noise figure from 0.5 to 3 GHz (datasheet: 0.3 dB at 1.9 GHz, OIP3 +39 dBm); 65 mA from 5 V

Qorvo QPL9547: a GaAs pHEMT LNA, internally matched to 50 ohm, 0.1–6 GHz. It sets the noise figure of every receiver here, so put it first (before the filter) unless a strong neighbour overloads it; +39 dBm OIP3 makes that rare. The 18 nH wire–wound choke (Qorvo's value) feeds 5 V into the output pin; 3.3 k from 5 V to Vbias sets 65 mA. Never cable a transmitter into it without 30 dB of attenuation (22 dBm absolute maximum at the input).

X8 GNSS front end, MAX2659

cell (3,0)

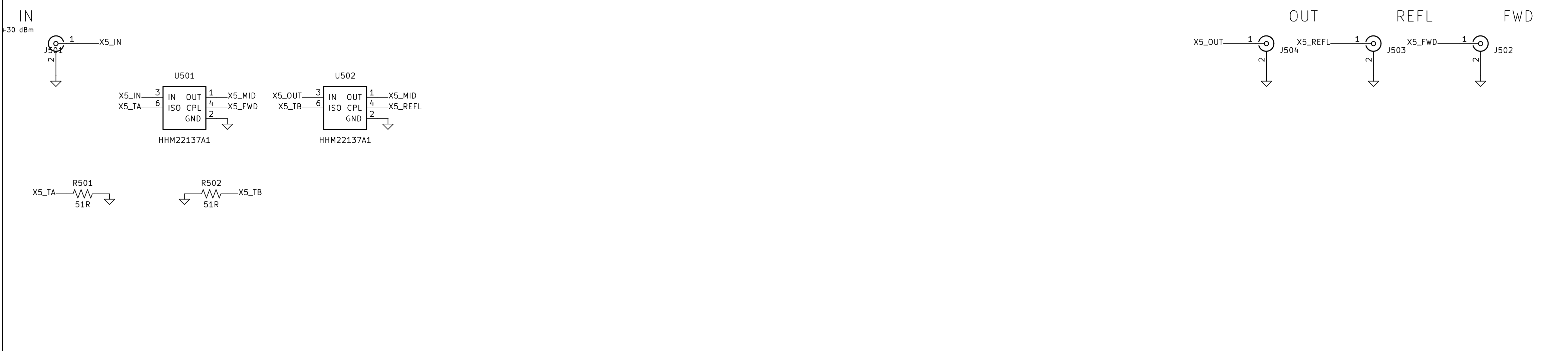


EXPECT: about 19 dB of gain from 1559 to 1606 MHz (GPS L1, Galileo E1, BeiDou B1, GLONASS L1), 1.5 dB noise figure with the first SAW, 50 dB or more of rejection at 900 MHz and 2.4 GHz; 3.3 V for an active antenna on ANT

SAW – LNA – SAW, the front end of every GNSS receiver. Murata's SAFFB1G56 passes 1561, 1575 and 1602 MHz with 1.6 dB of loss; Maxim's MAX2659 gives 20.5 dB with a 0.8 dB noise figure at 4 mA, its output matched and DC–blocked inside, its input matched by 470 pF and 6.8 nH (the datasheet's values). The bias–T feeds 3.3 V through 10 ohm and 18 nH to an active antenna's LNA (10–30 mA). OUT to X2's RF IN with X3 at 1575.42 MHz: the satellites' codes at the ADCs.

X5 coupler, 2 x HHM22137A1

cells (0,1), (1,1)

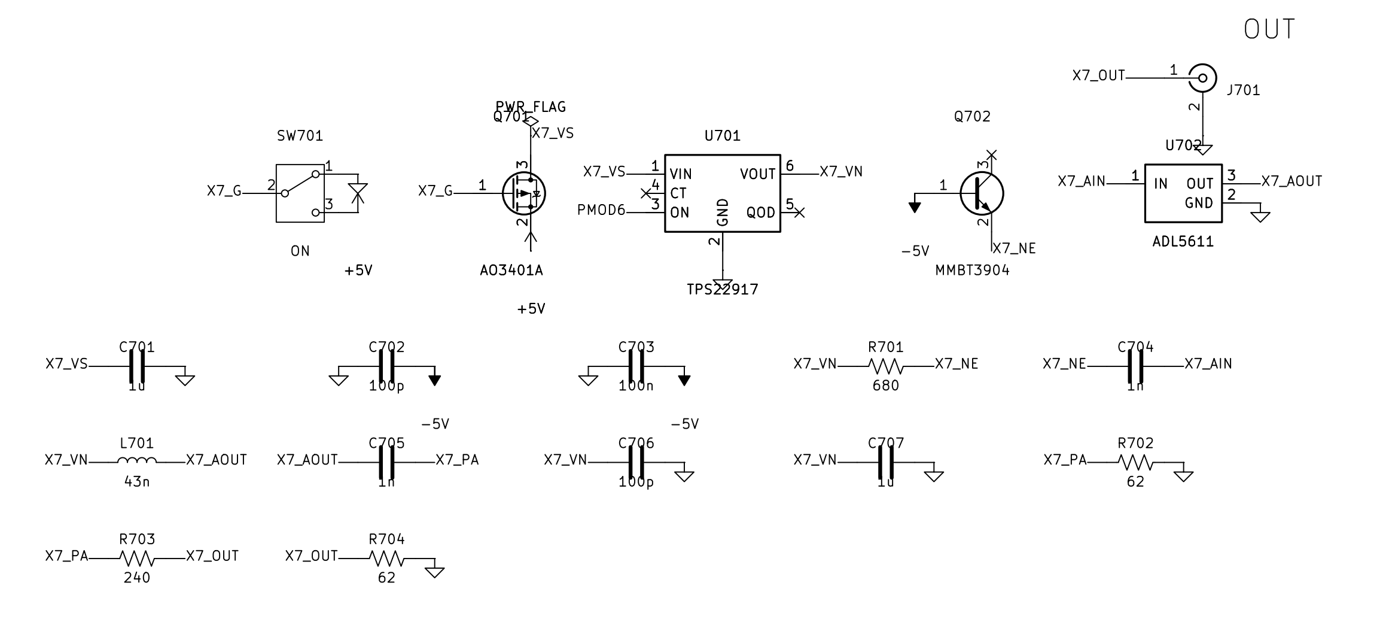


EXPECT: FWD 25 dB below the wave going from IN to OUT and REFL 25 dB below the wave coming back from OUT, 0.6–2.7 GHz (27 dB at 0.6, 14 dB at 3.5 GHz); 0.3 dB through; about 20 dB of directivity: with a device on OUT, REFL / FWD is its reflection coefficient

Two TDK HHM22137A1 directional couplers back to back, each with its isolated port in 51 ohm: the first samples the wave going from IN to OUT (FWD), the second, turned round, the wave coming back from OUT (REFL). X1 on IN, a device on OUT, REFL into X2 (and FWD into X6 or through X11): the device's S11 in magnitude and phase. Directivity (about 20 dB) limits the smallest reflection it sees; calibrate with open, short and load. +35 dBm through; terminate an unused port in 50 ohm.

X7 noise source, ADL5611

cell (2,1)

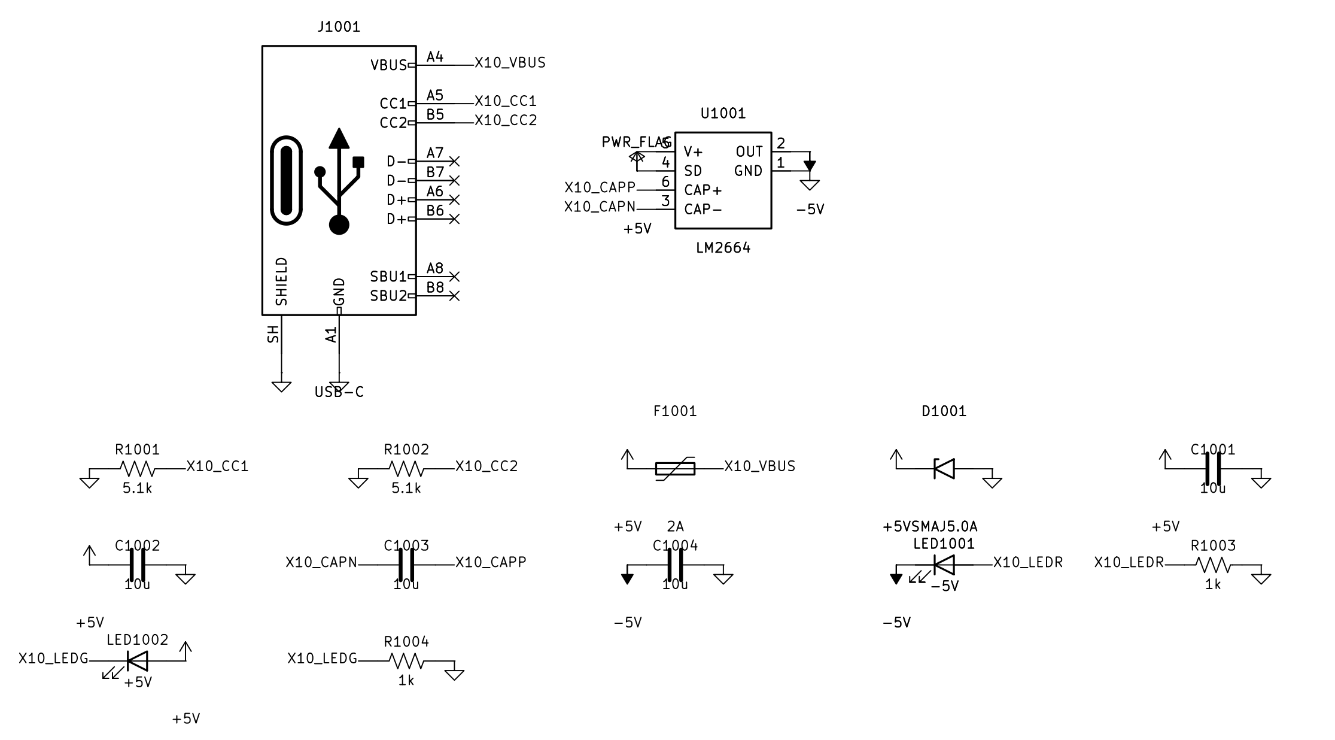


EXPECT: IO6 high: a flat noise spectrum from 50 MHz to 4 GHz, roughly 15–30 dB ENR at OUT (measure it); IO6 low: a 50 ohm load at room temperature; 20 dB of pad keeps OUT matched in both states

Avalanche noise: the MMBT3904's base–emitter junction breaks down near 8 V (6 V minimum), its base on –5 V and its emitter fed from the switched 5 V through 680 ohm. The ADL5611 (22 dB, 43 nH choke, ADI's values) lifts it; the 20 dB pad makes the output 50 ohm whether the amplifier is on or off, so the cold state is a room–temperature termination. Calibrate the ENR against a 50 ohm load in ice water and in boiling water (273 and 373 K) or in liquid nitrogen (77 K).

X10 power, USB–C 5 V in: +5, –5 out

cell (3,1)

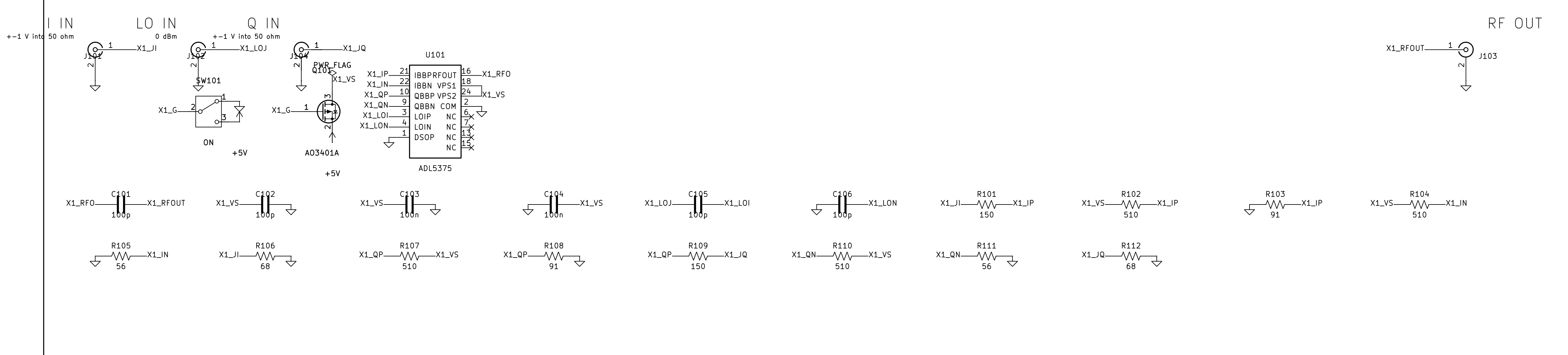


EXPECT: +5 V from the USB–C (up to 1.5 A), –5 V at 40 mA; the green LED says 5 V is in, the red one that –5 V is up

USB–C power only: 5.1 k on each CC asks any source for 5 V. A 2 A PTC, a 5 V TVS (clamps at 9.2 V) and 20 uF. The LM2664 charge pump makes –5 V for X2's difference amplifiers and X7's noise junction (40 mA available, 25 used). Every powered tile switches its own +5 V with a slide switch and a P–FET, and regulates its own 3.3 V where it needs one.

X1 I/Q modulator, ADL5375

cells (0,2), (1,2)

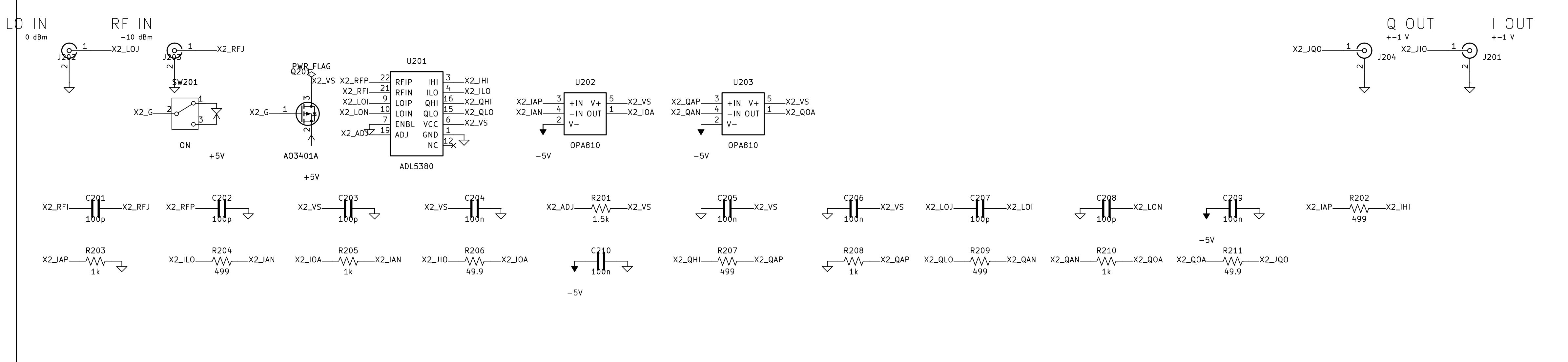


EXPECT: about 0 dBm of single–sideband RF at full scale (I and Q +–1 V at 1 V p–p differential equivalent), 0.4–6 GHz, sideband and carrier suppression near 40 dB untrimmed; baseband flat to 95 MHz, far past the DACs' 12 MHz

ADI ADL5375–05: two Gilbert–cell mixers and a quadrature LO splitter, 0.4–6 GHz, baseband bias 0.5 V. Each input is single–ended from a DAC: 68 ohm at the jack, 150 ohm into 510 / 91 ohm from 5 V (0.50 V, gain 0.34); the other pin of each pair sits at 0.50 V on 510 / 56 ohm. LO in single–ended through 100 pF (0 dBm; –6 to +6 dBm works); RF out is 50 ohm through 100 pF. Trim the carrier with small DC offsets on I and Q, the sideband with the DACs' gain and phase.

X2 I/Q demodulator, ADL5380

cells (2,2), (3,2)

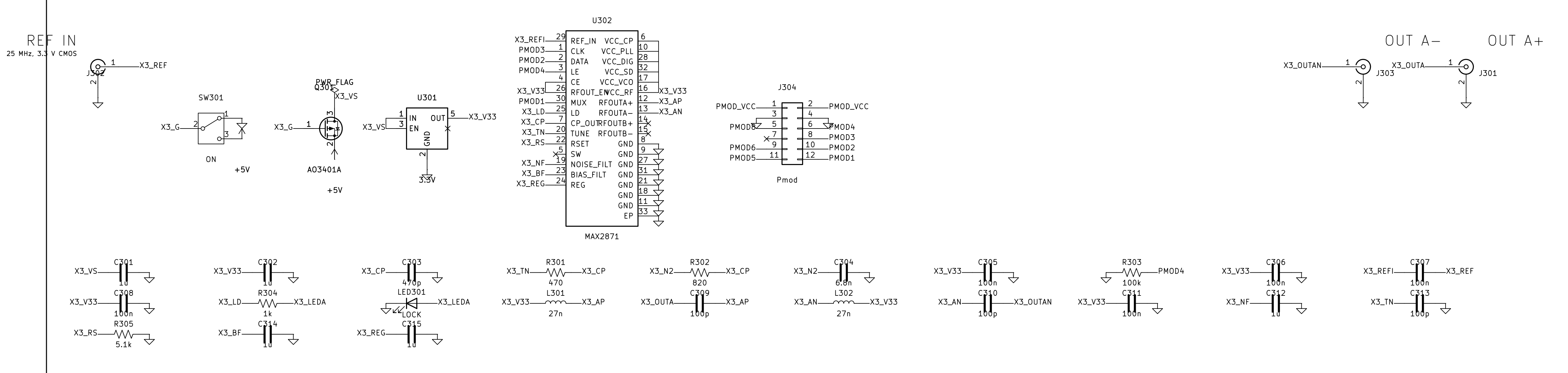


EXPECT: I and Q of the RF input against the LO, 0.4–6 GHz: about 7 dB of conversion gain from the chip, x2 in the amplifiers, +–1 V at the ADC for about –4 dBm of RF; flat from DC to beyond the ADCs' 23 MHz

ADI ADL5380: a quadrature demodulator, 0.4–6 GHz, 390 MHz of baseband, I and Q as differential pairs at 2.5 V. Two OPA810 difference amplifiers (499 ohm in, 1 k feedback and to ground: gain 2) make each pair one output about 0 V from +5 / –5 V; 49.9 ohm out. RF and LO single–ended through 100 pF, the other pin to ground through 100 pF (a balun would gain 1–3 dB and IP2). ADJ 1.5 k to 5 V suits 0.4–3 GHz (open it above 5 GHz). With X3's two outputs on the LOs: one coherent radio.

X3 synthesizer, MAX2871

cells (0,3), (1,3)

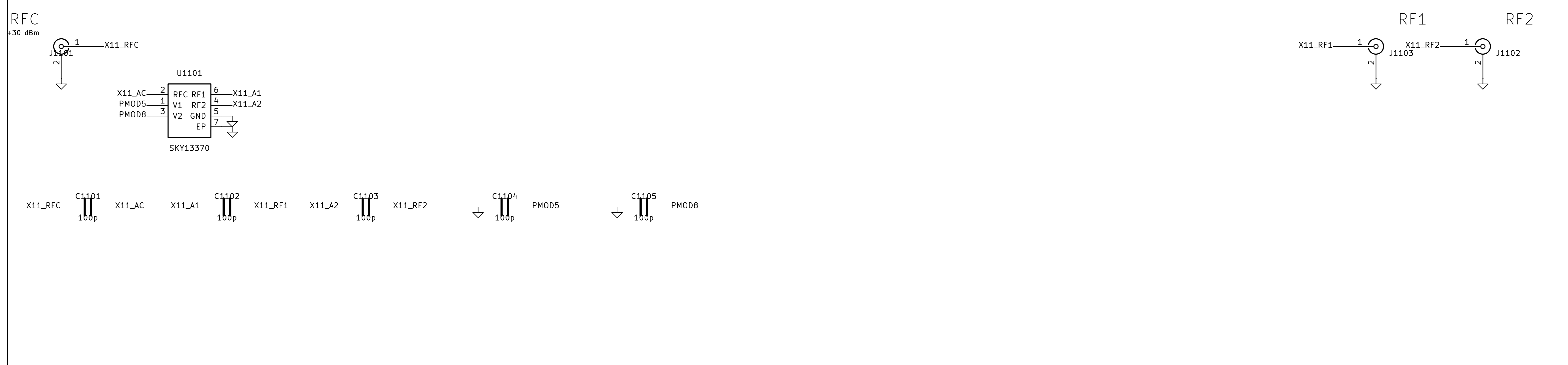


EXPECT: any frequency from 23.5 MHz to 6 GHz on OUT A+ and OUT A– (each –4 to +5 dBm, 180 degrees apart), locked to the 25 MHz on REF IN (the ECP5 board's CLK OUT or RF–MOD's REF OUT): OUT A+ is X1's LO and OUT A– X2's, one phase for both

MAX2871 as CostasLink built it: fractional–N PLL with four VCOs, programmed over SPI from the Pmod (SCLK, SDATA, LE; MUXOUT back). The loop filter is third order, 80–120 kHz with 61 degrees of margin for both 915 MHz (VCO / 4) and 2.4 GHz (VCO / 2) at ICP 2.56 mA (RSET 5.1 k, CP = 7). RFOUTA+ and RFOUTA– each feed a jack through 27 nH to 3.3 V and 100 pF: two outputs, 180 degrees apart; put a 50 ohm terminator on one that is unused. RFOUTB is open and must stay disabled. CE and RFOUT_EN are tied high, LE has 100 k down. REF IN is unterminated (3.3 V CMOS): the ECP5 board's CLK OUT or RF–MOD's REF OUT. OUT A+ into X1's LO and OUT A– into X2's at –1 dBm (X1 wants 0, X2 –6 to +6).

X11 switch, SKY13370

cell (2,3), (3,3)



EXPECT: 0.7 dB through, 31 dB of isolation at 2.5 GHz (0.5–6 GHz); the off throw is a 50 ohm load; switches in about 100 ns

Skyworks SKY13370: an absorptive SPDT, both throws matched in every state, so a Dicke radiometer's reference (X7 off: a room–temperature load) and its antenna see the same receiver. V1 high and V2 low connects RFC to RF1; the reverse to RF2. Pmod IO5 drives V1 and IO8 V2 from the FPGA, so the switching is synchronous with the ADC's samples. A T/R switch for one antenna, an antenna switch for two, a Dicke switch, a pulse gate.

GND GND GND

no mounting holes: four stick–on rubber feet, marked on the back

ground loops for the scope's clips, one per channel, in the board's margins

PWR_FLAG

Drawn by scripts/gen_sch.py from boards/rf_mod_2.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
an I/Q modulator and demodulator, a synthesizer and splitter, an LNA, a log detector, a switch, a noi
Board: Jason Gallicchio and Claude | open hardware, CERN–OHL–P–2.0 | github.com/gallicchio/
Sheet: /
File: rf_mod_2.kicad_sch
Title: RF–MOD–2: coherent I/Q and microwave physics
Size: A1 Date: 2026–10–25 **Rev: A**
KiCad E.D.A. 10.0.7 Id: 1/1